



HSBC x BAIF HACKATHON DELIVERABLE

अनुवादिनी

Breaking language barriers in rural India. A comprehensive, offline localization engine translating text, audio, and video manuals instantly.

Offline execution

 Local CPU processing

 Text • Audio • Video

THE FIELD CHALLENGE VS BAIF NEEDS

⚠ The Rural Reality

- 🚫 **Network Disconnection:** Internet is highly unreliable or non-existent in active sub-districts and field schools.
- 💰 **Restrictive API Overhead:** Standard cloud API models (OpenAI, DeepL) incur compounding transaction and subscription fees.
- 🔗 **Tool Fragmentation:** Translating a complete instruction kit currently requires disjointed software for transcription, synthesis, and subtitling.
- ⚙ **Corporate Data Exposure:** Proprietary instructional intellectual property is leaked outside of client environments when sent to third-party translation servers.

✅ What BAIF Needed

- ✅ **100% Offline execution:** Zero active external internet ping needed during runtime.
- ✅ **Multi-format Support:** Text transcription, Voiceovers, and Embedded Subtitles within a single application.
- ✅ **Intel CPU Optimization:** Capable of performing local ASR and NMT models on existing BAIF single-core computers.
- ✅ **Regional Preservation:** Specialized grammatical models tuned to natural conversational Hindi and Marathi dialects.

SOLUTION ARCHITECTURE



1. Data Ingestion

1

Unified File Intake

Accepts .mp4, .mov, .mp3, .wav, and .txt files up to 2 GB via Streamlit frontend.

2

Demultiplexing

Utilizes bundled local FFmpeg bin tools to extract and normalize audio frequencies instantly.



2. Core Processing

3

Speech Recog. & LID

Faster-Whisper processes waves on a local thread loop, executing automatic language detection.

4

NMT Translation

NLLB-200 translates complete transcription context, preventing paragraph fragment errors.



3. Synthesis & Output

5

Vocal Overlays & SRT

eSpeak NG generates dialect voice tracks while ASS engine aligns scaled subtitles.

6

Multiplexed Video

Composited ASS font scripts and voices are embedded into a final output via FFmpeg.

CORE OPEN-SOURCE ARCHITECTURE STACK



Faster-Whisper



NLLB-200



eSpeak NG



FFmpeg



FastAPI



Streamlit

HOW अनुवादिनी MEETS HACKATHON PILLARS

● KEY DIFFERENTIATORS

Evaluation Pillar	Our Implementation Strength
 IMPACT	Enables critical, direct, and multi-dialect educational manuals to easily reach rural Hindi- and Marathi-speaking communities.
 SCALABILITY	Modular FastAPI core prepared to integrate cluster chunking, batch operations, and future GPU queue support without changing processing services.
 READINESS	Handles automated CPU safety fallback, large files up to 2 GB, and cross-format processing (MP4, MKV, MP3, WAV, AAC, and TXT).
 USABILITY	Clean, drag-and-drop Streamlit UI. Offers instantaneous translation preview panels, dynamic subtitle track editing, and single-click outputs.
 SAFETY	100% offline local processing. Absolutely zero user inputs, transcripts, or field manuals leave your local corporate networks.
 INCLUSIVITY	Ensures robust accessibility through both visual (dynamic subtitles) and physical (AI audio overlays) regional language generation.
 ZERO COST	Built strictly upon open-source model weights. Completely free of perpetual licensing overhead or recurring cloud subscription bills.

CORE TECHNICAL EDGE & LIVE DEMO

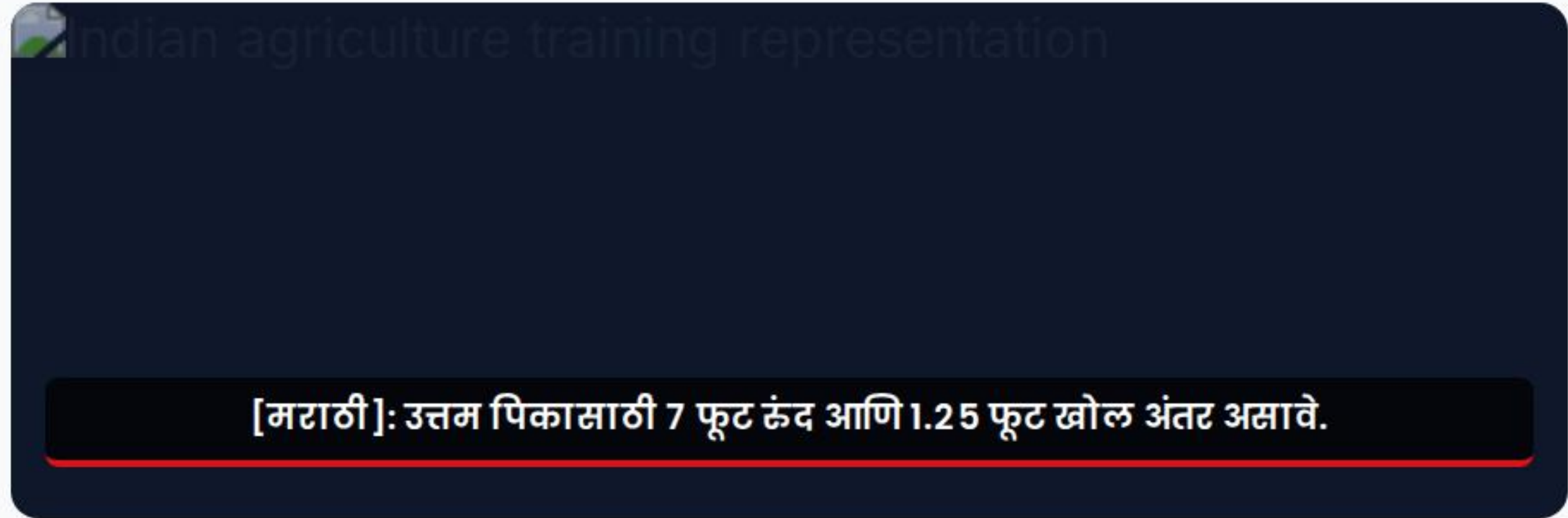
● LIVE EXECUTION

- ✓ Full-Context Translation
- ✓ ASS Subtitle Scaling
- ✓ Devnagari Font Bundling
- ✓ Subtitle Speed Sync
- ✓ Venv Portability
- ✓ 2 GB Streamlit Limit

"अनुवादिनी transforms multilingual educational content into accessible, localized, and practical knowledge—anytime, anywhere, completely independent of the internet."

Translation Viewer

http://localhost:8501/outputs



[मराठी]: उत्तम पिकासाठी 7 फूट रुंद आणि 1.25 फूट खोल अंतर असावे.

eSpeak Synthesis Overlay (Marathi)



| IMAGE SOURCES



https://images.splitshire.com/full/Ramu-the-Farmer:-A-Kid-Friendly-Animated-Village-Morning_Prre3.png

Source: www.splitshire.com